E201 Particle Nature of Light

E2.1.1 · E2.1.4 · E2.1.2 · E2.1.3 Photoelectric emission and stopping potential

E2.1.1 · E2.1.4 The photoelectric effect and threshold frequency

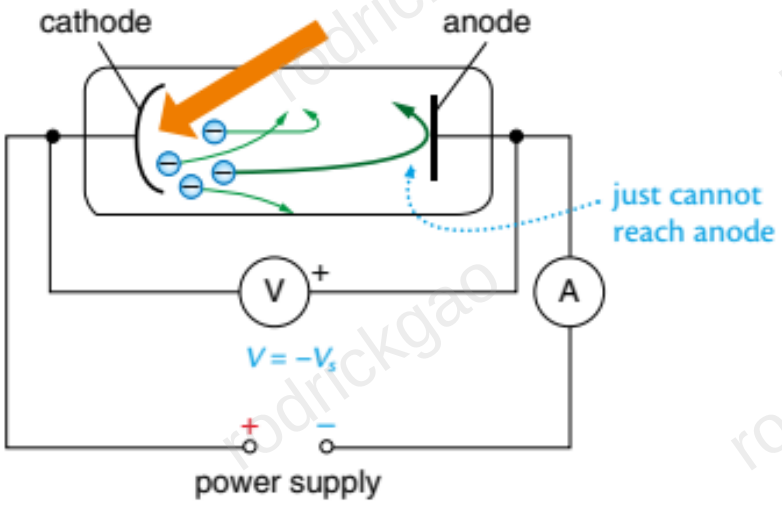
Photoelectrons can be emitted when an incident photon's energy overcomes the energy required for escape from a surface. Each surface has a threshold frequency. Below it, increasing ordinary light intensity cannot cause single-photon emission. Above threshold, increasing intensity chiefly increases the number of emitted electrons.

E2.1.2 · E2.1.3 Stopping potential

Stopping potential is the minimum reverse-voltage magnitude that just prevents even the most energetic photoelectrons from reaching the collector, reducing photocurrent to zero. It measures maximum, not average, kinetic energy. Use positive potential and elementary-charge magnitudes in the energy relation; reverse voltage does not mean negative kinetic energy.

$$K_{\max} = eV_s$$

$K_{\max}$: maximum photoelectron kinetic energy (J); V_s: stopping potential magnitude (V); e = 1.60 × 10^{-19} C, positive elementary charge.



Reverse voltage prevents photoelectrons reaching the collector; at the stopping potential the photocurrent just reaches zero.

Textbook sources: E201 · PDF 2, 5, 6, 9

E2.1.5 Photon energy

Light exchanges energy in photons. One photon's energy is proportional to frequency and inversely proportional to vacuum wavelength. Increasing frequency increases energy per photon; increasing intensity at fixed frequency mainly increases photon rate. With SI Planck constant, use wavelength in metres to obtain joules, then convert to eV if needed.

$$E = hf = \frac{hc}{\lambda}$$

Photon energy E (J); h = 6.63 × 10^{-34} J s; f: Hz; c ≈ 3.00 × 10^{8} m s^{-1}; λ: vacuum wavelength (m).

Textbook sources: E201 · PDF 14

E2.1.6 · E2.1.7 Work function and threshold frequency

Work function is the minimum energy needed to free an electron from a surface, depending on material and surface condition. A threshold-frequency photon just supplies that energy, so maximum emitted kinetic energy tends to zero. Keep work function, photon energy and kinetic energy consistently in joules or electronvolts.

$$\phi = hf_0$$

φ: work function (J); f_0: threshold frequency (Hz); h = 6.63 × 10⁻³⁴ J s. Use consistent energy units.

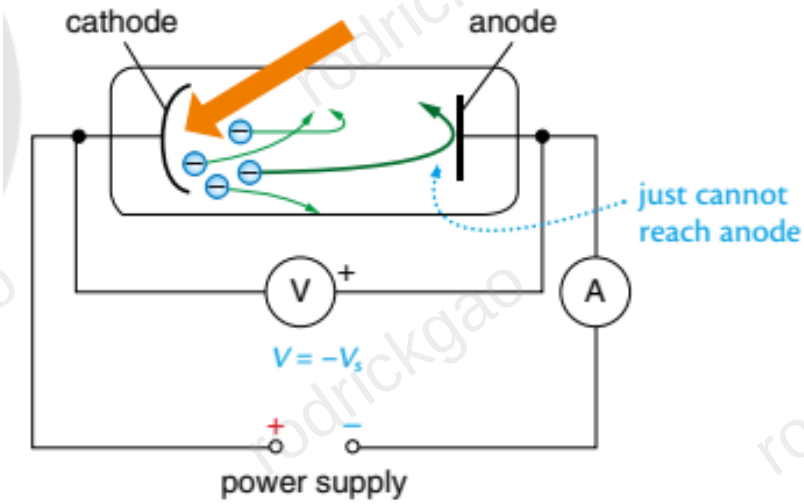
Textbook sources: E201 · PDF 18

E2.1.8 Einstein's photoelectric equation

A photon's energy first overcomes the work function; the remainder is the maximum photoelectron kinetic energy. Higher frequency increases this maximum, whereas higher intensity at fixed frequency does not. A negative calculated result means insufficient photon energy and no emission, not electrons with negative kinetic energy.

$$K_{\max} = hf - \phi = h(f - f_0)$$

Photoemission requires $f \geq f_0$. $K_{\max}$ and φ: J; f, f_0: Hz; h = 6.63 × 10⁻³⁴ J s. A negative result means no photoemission, not negative kinetic energy.



Reverse voltage prevents photoelectrons reaching the collector; at the stopping potential the photocurrent just reaches zero.

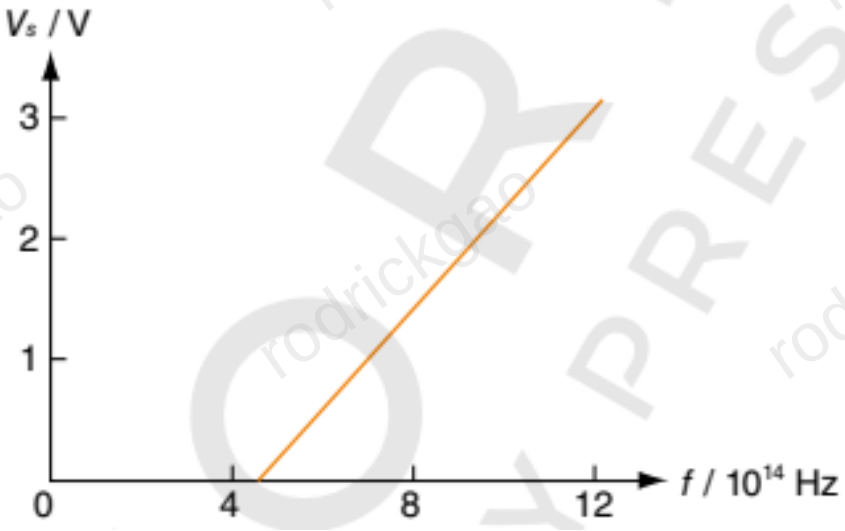
Textbook sources: E201 · PDF 5, 20

E2.1.9 Stopping-potential-frequency graphs

Above threshold, stopping potential varies linearly with frequency for a fixed material. The slope determines Planck's constant and the frequency-axis intercept gives threshold frequency. The extrapolated negative-potential region is mathematical only, not emission below threshold. Changing material changes intercepts, while the ideal-model slope stays unchanged.

$$V_s = \frac{h}{e} f - \frac{\phi}{e}$$

For $f \geq f_0$: V_s is stopping-potential magnitude (V); slope h/e; intercept −φ/e. h = 6.63 × 10⁻³⁴ J s; e = 1.60 × 10⁻¹⁹ C; φ: J.



Stopping potential varies linearly with frequency in the region where photoelectric emission occurs.

Textbook sources: E201 · PDF 26

E2.1.10 Wave-particle duality of light

Interference and diffraction are described by light's wave behaviour, while discrete energy transfers in the photoelectric effect require photons. These descriptions address different experimental features, not arbitrary switching between ordinary balls and mechanical waves. Choose the model appropriate to the observed phenomenon.

Textbook sources: E201 · PDF 28